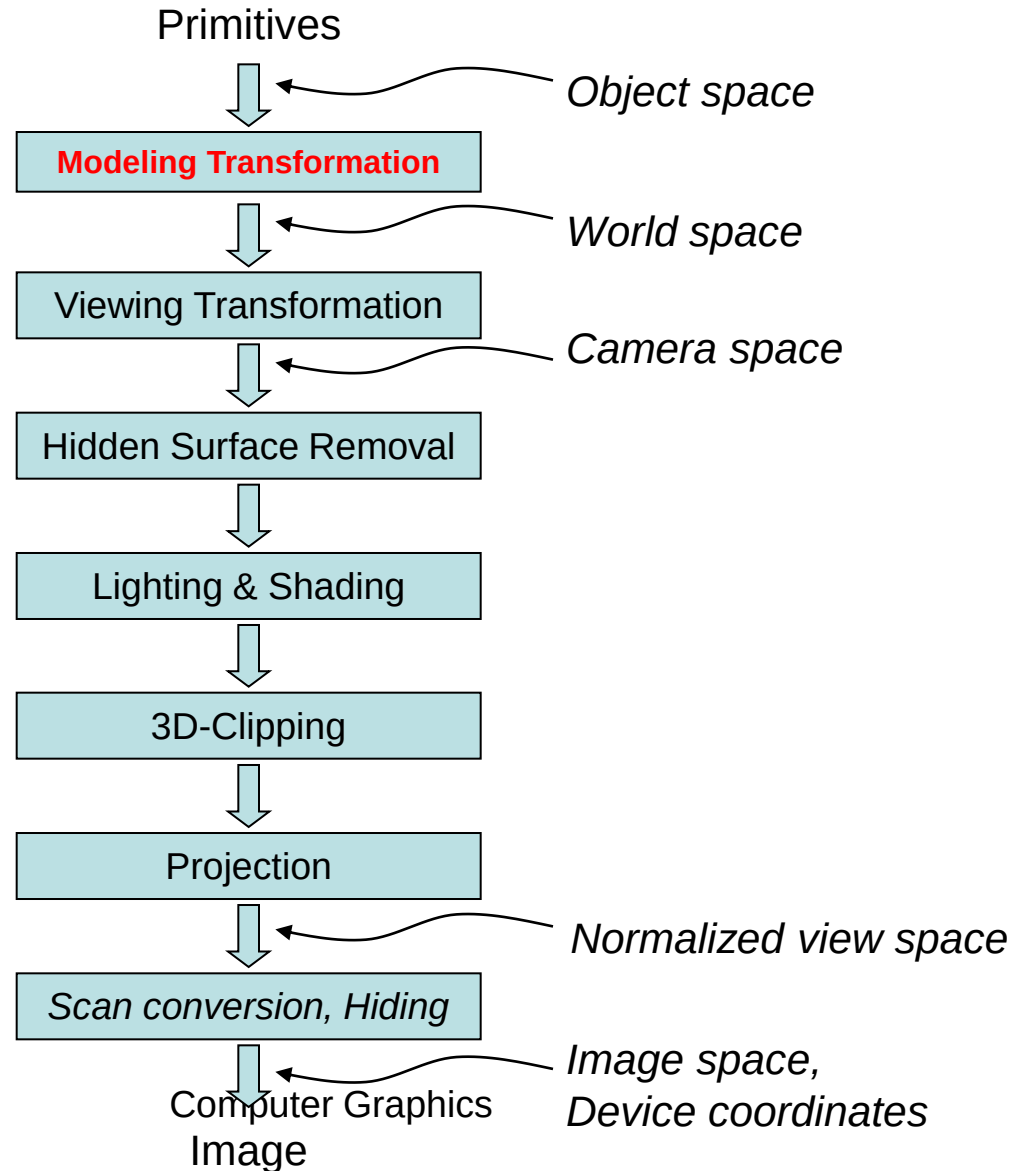


3D Viewing Pipeline

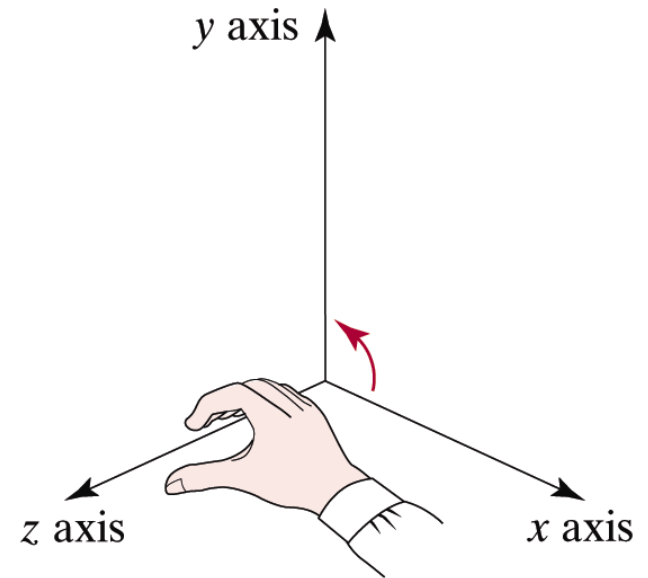
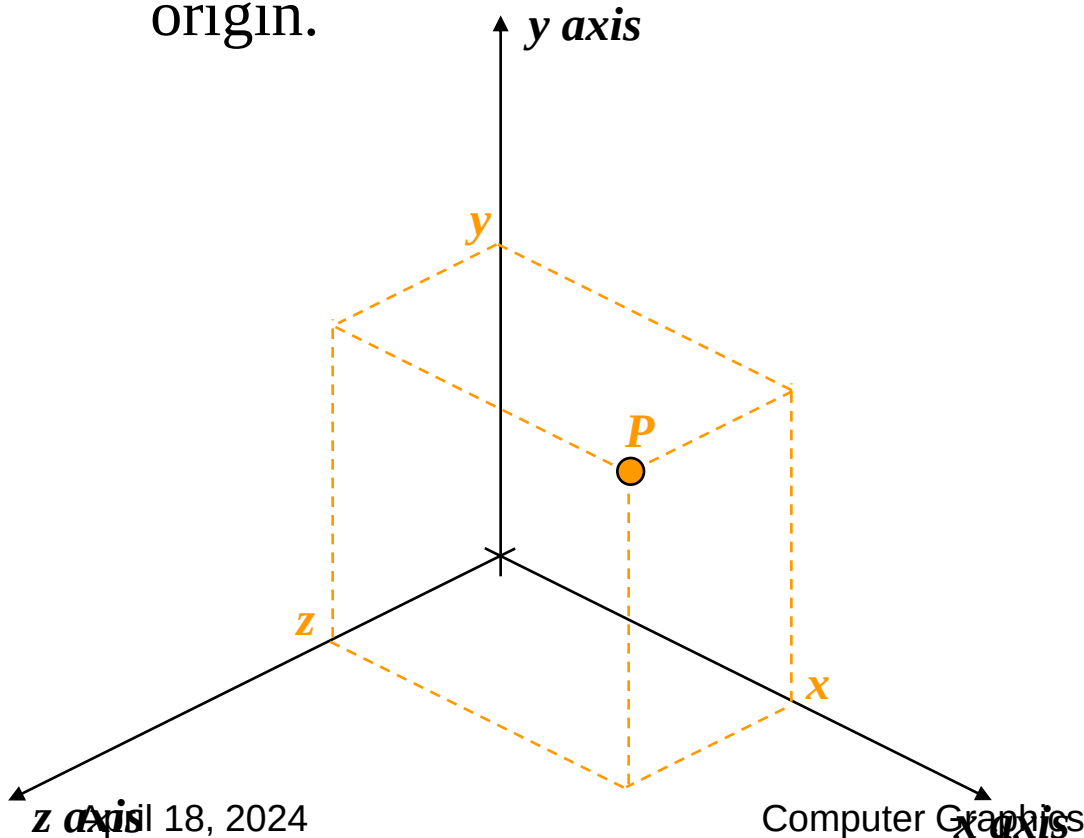


Contents

- 1. 3D Coordinate System**
2. Vector Fundamentals
3. 3D Homogeneous Coordinate
4. 3D Geometric Transformations
5. 3D Inverse Transformations
6. 3D Coordinate Transformations
7. Composite transformations

3D Coordinate System

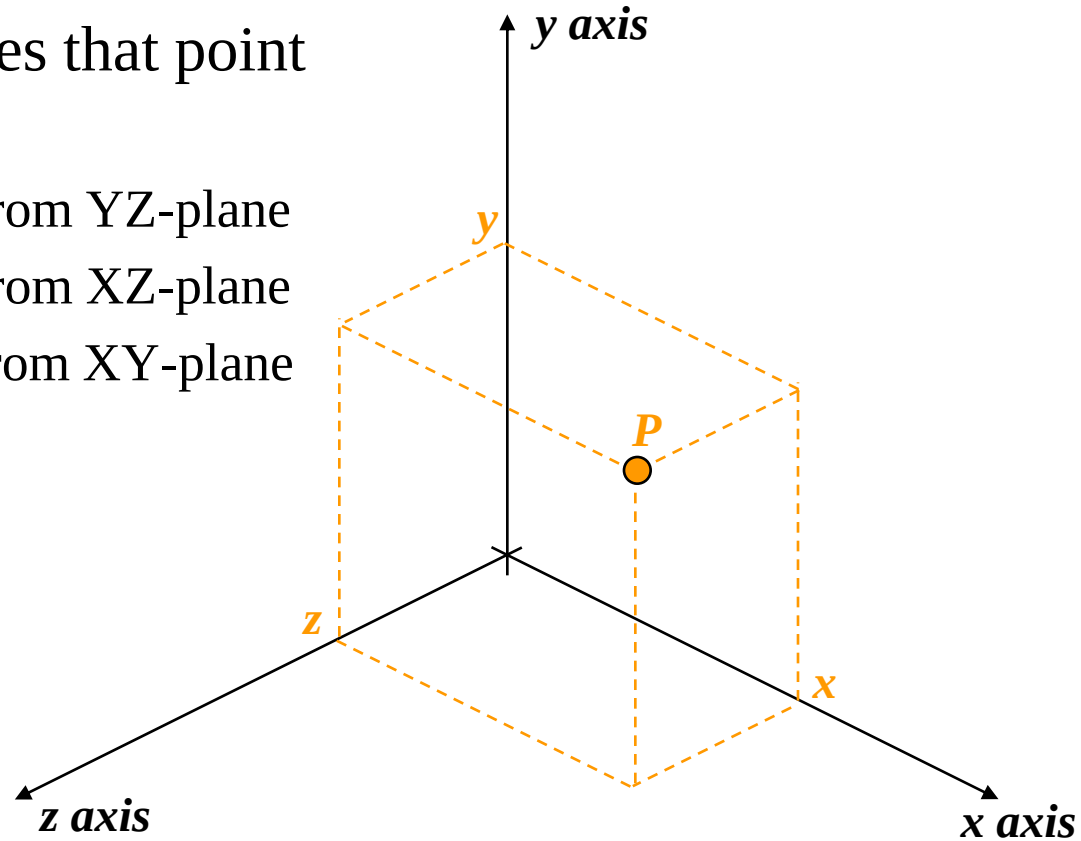
- 3D coordinate system consists of a reference point called origin and three mutually perpendicular passing through origin.



Right-Hand
Reference System

3D Coordinate System

- A point $P(x,y,z)$ specifies that point is
 - At a distance of x units from YZ -plane
 - At a distance of y units from XZ -plane
 - At a distance of z units from XY -plane

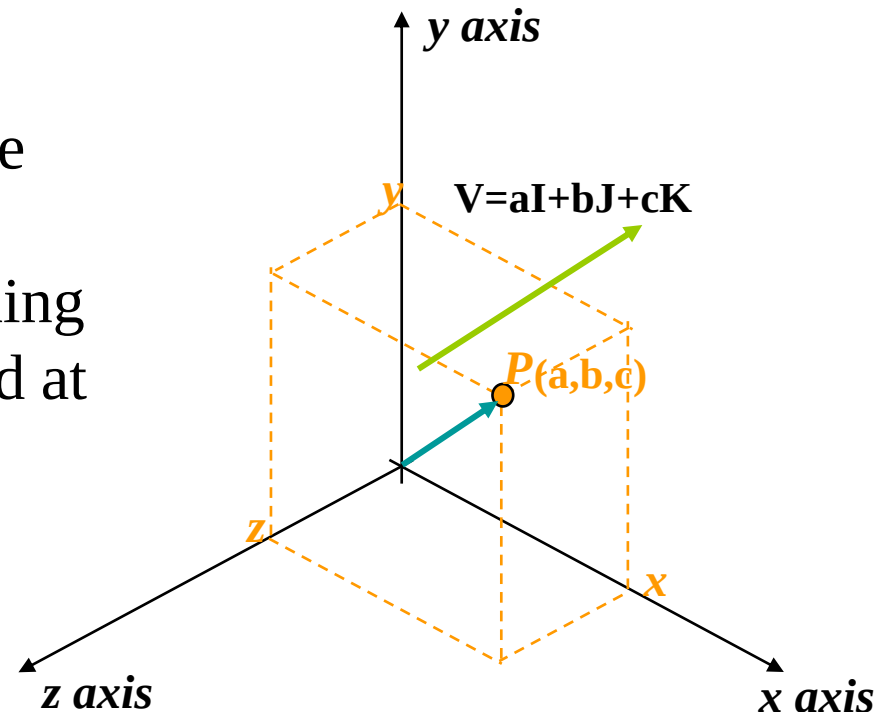


Contents

1. 3D Coordinate System
- 2. Vector Fundamentals**
3. 3D Homogeneous Coordinate
4. 3D Geometric Transformations
5. 3D Inverse Transformations
6. 3D Coordinate Transformations
7. Composite transformations

Vector Fundamentals

- In 3D there are 3 natural coordinate vectors I , J and K having unit along X , Y and Z axis respectively.
- Any vector $V = aI + bJ + cK$ can be resolved into three components (a, b, c) that represent corresponding point when the tail of V is placed at the origin.

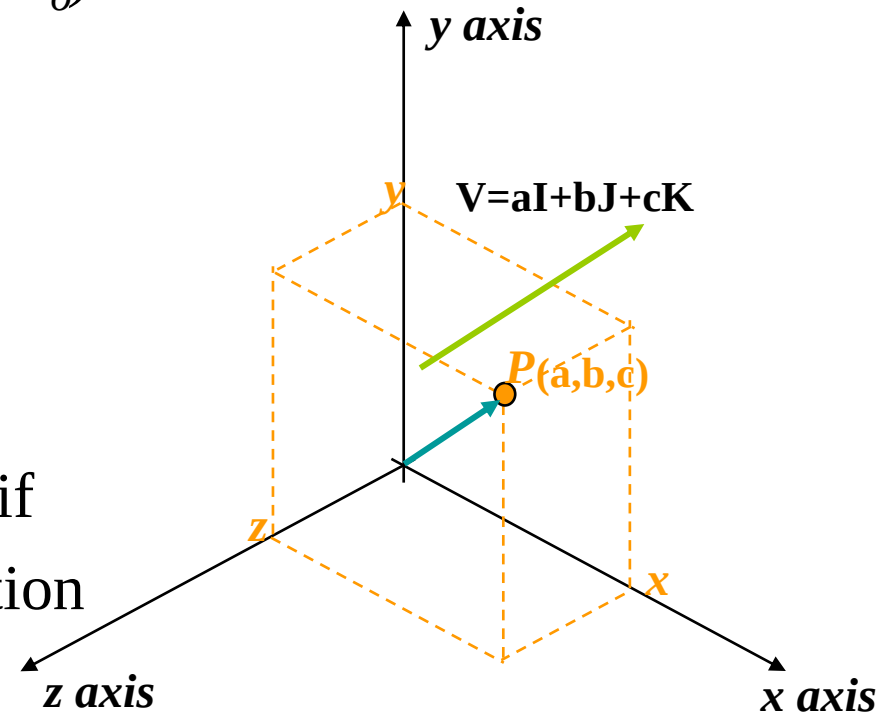
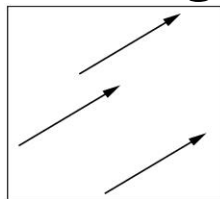


Vector Fundamentals

- If $P(x_0, y_0, z_0)$ and $Q(x_1, y_1, z_1)$ are two points in space then

$$\overrightarrow{PQ} = (x_1 - x_0)I + (y_1 - y_0)J + (z_1 - z_0)K$$

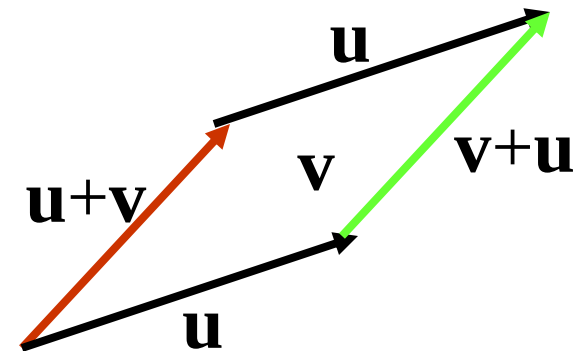
- Vector has two attributes:
 - **length**- a scalar denoted by $|v|$
 - **direction** in 3D space
- Equality: Two vectors are equal if they have same length and direction



Vector Fundamentals

1. **Vector Addition:** perform addition by the head-to-tail rule.

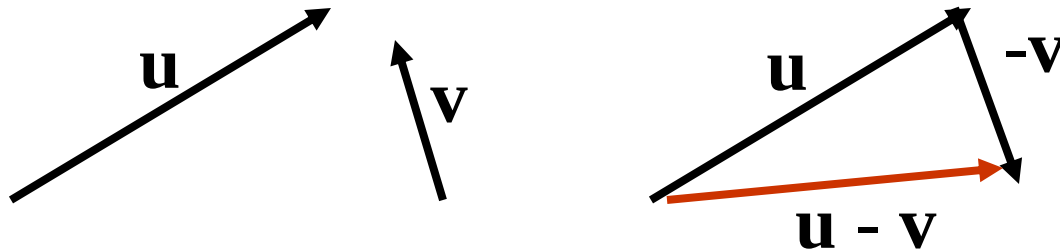
Note $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$



Vector Fundamentals

2. **Vector Subtraction:** To perform subtraction, given $\mathbf{u}$ and $\mathbf{v}$, we define $\mathbf{u} - \mathbf{v}$ to be $\mathbf{u} + (-\mathbf{v})$:

Note $\mathbf{u} - \mathbf{v} \neq \mathbf{v} - \mathbf{u}$



Vector Fundamentals

3. **Zero Vector:** i.e. vector with zero length and no direction.

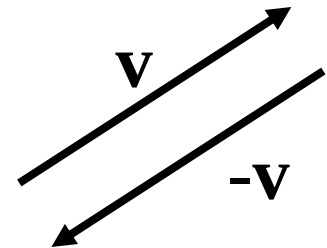
$$\mathbf{v} + (-\mathbf{v}) = \vec{\mathbf{0}}$$

4. **Magnitude of Vector** of a vector $\mathbf{u} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$ is a scalar quantity defined as:

$$|\mathbf{u}| = \sqrt{a^2 + b^2 + c^2}$$

5. **Unit vector** for any vector $\mathbf{u}$ is a vector having direction of $\mathbf{u}$ and unit magnitude. It can be found as

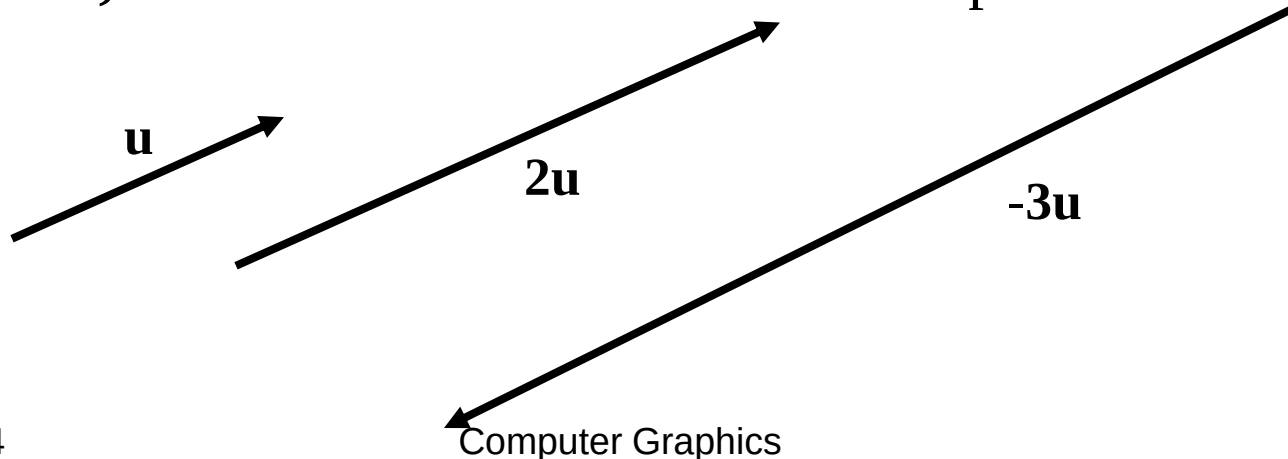
$$\mathbf{u}_v = \frac{\mathbf{u}}{|\mathbf{u}|}$$



Vector Fundamentals

6. **Scalar Multiplication:** (Scalar times a vector) Let α be a scalar and $\mathbf{u}$ a vector we define $\alpha\mathbf{u}$ to be the vector with **length** $|\alpha| |\mathbf{u}|$ with
- the **direction** of $\mathbf{u}$ if $\alpha > 0$
 - the opposite of $\mathbf{u}$ if $\alpha < 0$

if $\alpha = 0$, the vector is the zero vector or a point.



Vector Fundamentals

7. **Dot Product:** Given two vectors $\mathbf{u} = a_1\mathbf{I} + b_1\mathbf{J} + c_1\mathbf{K}$ and $\mathbf{v} = a_2\mathbf{I} + b_2\mathbf{J} + c_2\mathbf{K}$ the dot product of the vectors is defined to be the *scalar* as follows

$$\mathbf{u} \cdot \mathbf{v} = a_1 \cdot a_2 + b_1 \cdot b_2 + c_1 \cdot c_2$$

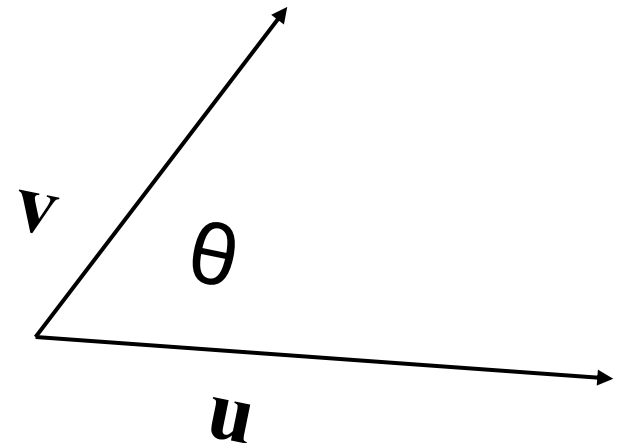
$$\text{or } \mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta, 0 \leq \theta \leq \pi$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$

Also

$$\cos \theta = \mathbf{u} \cdot \mathbf{v} / |\mathbf{u}| |\mathbf{v}|$$

$$\text{Or } \theta = \cos^{-1}(\mathbf{u} \cdot \mathbf{v} / |\mathbf{u}| |\mathbf{v}|)$$



Vector Fundamentals

The **dot product** is a scalar value that tells us something about the relationship between two vectors

- If $\mathbf{u} \cdot \mathbf{v} > 0$ then $0 < \theta < 90^\circ$
 - Vectors point in the same general direction
- If $\mathbf{u} \cdot \mathbf{v} < 0$ then $\theta > 90^\circ$
 - Vectors point in opposite direction
- If $\mathbf{u} \cdot \mathbf{v} = 0$ then $\theta = 90^\circ$
 - Vectors are perpendicular
 - (or one or both of the vectors is degenerate (0,0,0))

Vector Fundamentals

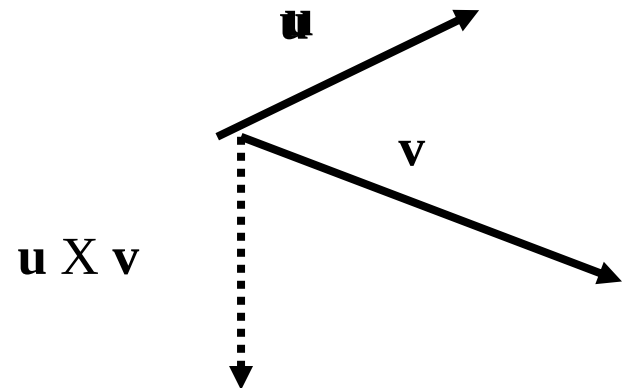
8. **Cross Product:** Given two vectors $\mathbf{u} = a_1\mathbf{I} + b_1\mathbf{J} + c_1\mathbf{K}$ and $\mathbf{v} = a_2\mathbf{I} + b_2\mathbf{J} + c_2\mathbf{K}$ the cross product of the vectors is defined to be the new *vector* whose length is $|\mathbf{u}||\mathbf{v}| \sin \theta$, $0 \leq \theta \leq \pi$ and direction is given by the right hand rule.

$$\mathbf{u} \times \mathbf{v} = (b_1 \cdot c_2 - c_1 \cdot b_2)\mathbf{I} + (c_1 \cdot a_2 - a_1 \cdot c_2)\mathbf{J} + (a_1 \cdot b_2 - b_1 \cdot a_2)\mathbf{K}$$

or $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$, $0 \leq \theta \leq \pi$ where θ is the angle b/w $\mathbf{u}$ & $\mathbf{v}$

$$\sin \theta = |\mathbf{u} \times \mathbf{v}| / |\mathbf{u}| |\mathbf{v}|$$

$$\text{Or } \theta = \sin^{-1}(|\mathbf{u} \times \mathbf{v}| / |\mathbf{u}| |\mathbf{v}|)$$



Vector Fundamentals

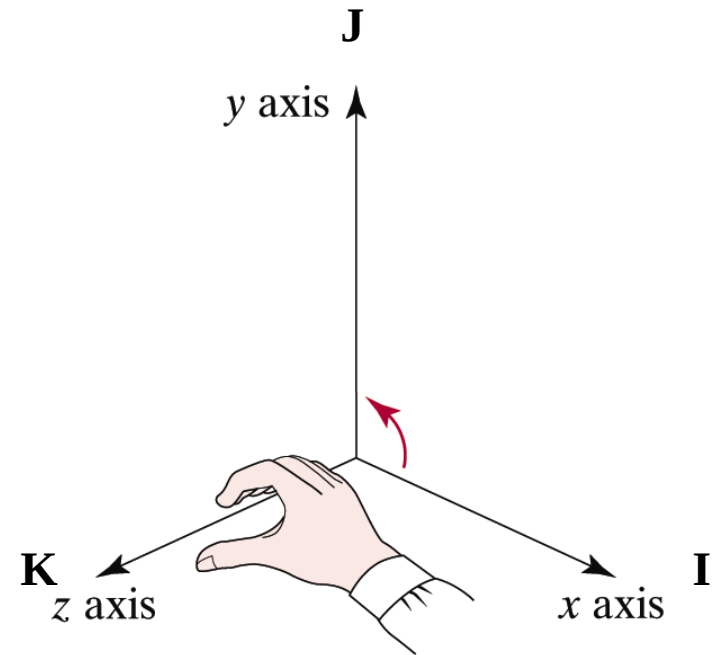
- In a Right handed coordinate System

- $I \times J = K$

- $J \times K = I$

- $I \times K = -J$

- or $K \times I = J$



Right-Hand
Reference System

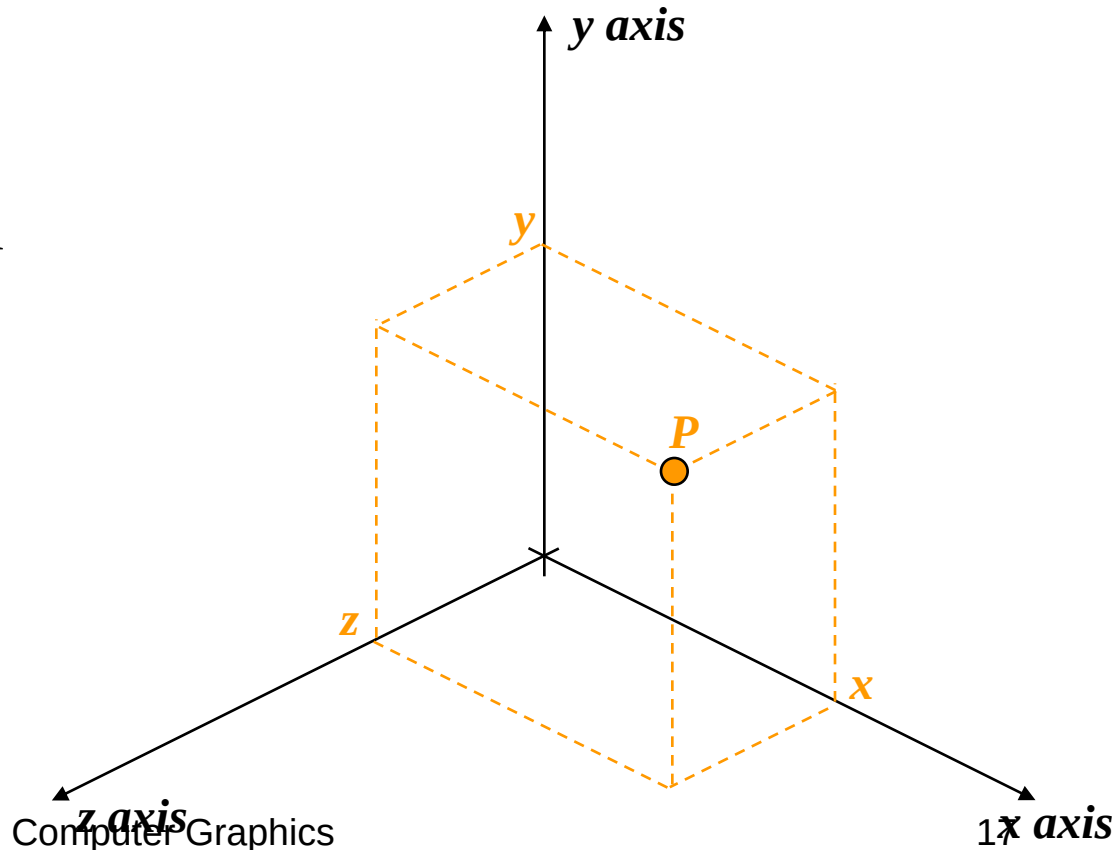
Contents

1. 3D Coordinate System
2. Vector Fundamentals
- 3. 3D Homogeneous Coordinate**
4. 3D Geometric Transformations
5. 3D Inverse Transformations
6. 3D Coordinate Transformations
7. Composite transformations

3D Homogeneous Coordinates

- Similar to the 2-D situation we can use homogeneous coordinates for 3-D transformations - 4 coordinate column vector
- All transformations can then be represented as matrices

$$P(x, y, z) = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Homogeneous Coordinates

- Transform a point:

$$\begin{bmatrix} p'_x \\ p'_y \\ p'_z \\ 1 \end{bmatrix} = \begin{bmatrix} m_{xx} & m_{xy} & m_{xz} & d_x \\ m_{yx} & m_{yy} & m_{yz} & d_y \\ m_{zx} & m_{zy} & m_{zz} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} m_{xx}p_x + m_{xy}p_y + m_{xz}p_z + d_x \\ m_{yx}p_x + m_{yy}p_y + m_{yz}p_z + d_y \\ m_{zx}p_x + m_{zy}p_y + m_{zz}p_z + d_z \\ 0 + 0 + 0 + 1 \end{bmatrix}$$
$$\mathbf{M} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} + \vec{d}$$

- Top three rows are the affine transform!
- Bottom row stays 1

3D Homogeneous Coordinates

- Transform a vector:

$$\begin{bmatrix} v'_x \\ v'_y \\ v'_z \\ 0 \end{bmatrix} = \begin{bmatrix} m_{xx} & m_{xy} & m_{xz} & d_x \\ m_{yx} & m_{yy} & m_{yz} & d_y \\ m_{zx} & m_{zy} & m_{zz} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \\ 0 \end{bmatrix} = \begin{bmatrix} m_{xx}v_x + m_{xy}v_y + m_{xz}v_z + 0 \\ m_{yx}v_x + m_{yy}v_y + m_{yz}v_z + 0 \\ m_{zx}v_x + m_{zy}v_y + m_{zz}v_z + 0 \\ 0 + 0 + 0 + 0 \end{bmatrix}$$


 $\mathbf{M} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$

- Top three rows are the linear transform
 - Displacement **d** is properly ignored
- Bottom row stays 0

3D Homogeneous Coordinates

- In Homogeneous arithmetic Legal operations always end in 0 or 1!

$$\text{vector+vector:} \quad \begin{bmatrix} \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 0 \end{bmatrix}$$

$$\text{vector-vector:} \quad \begin{bmatrix} \vdots \\ 0 \end{bmatrix} - \begin{bmatrix} \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 0 \end{bmatrix}$$

$$\text{scalar*vector:} \quad s \begin{bmatrix} \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 0 \end{bmatrix}$$

$$\text{point+vector:} \quad \begin{bmatrix} \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 1 \end{bmatrix}$$

$$\text{point-point:} \quad \begin{bmatrix} \vdots \\ 1 \end{bmatrix} - \begin{bmatrix} \vdots \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 0 \end{bmatrix}$$

$$\text{point+point:} \quad \begin{bmatrix} \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} \vdots \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 2 \end{bmatrix}$$

$$\text{scalar*point:} \quad s \begin{bmatrix} \vdots \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ s \end{bmatrix}$$

$$\left\{ \begin{array}{l} \text{weighted average} \\ \text{affine combination} \end{array} \right\} \text{ of points:} \quad \frac{1}{3} \begin{bmatrix} \vdots \\ 1 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} \vdots \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} \vdots \\ 1 \end{bmatrix}$$

3D Homogeneous Coordinates

- Rotation, Scale, and Translation of points and vectors unified in a single matrix transformation:

$$\mathbf{p}' = \mathbf{M} \mathbf{p} \quad \text{Where } \mathbf{M} = \begin{bmatrix} m_{xx} & m_{xy} & m_{xz} & d_x \\ m_{yx} & m_{yy} & m_{yz} & d_y \\ m_{zx} & m_{zy} & m_{zz} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Matrix has the form:
 - Last row always 0,0,0,1
- Transforms compose by matrix multiplication.

Contents

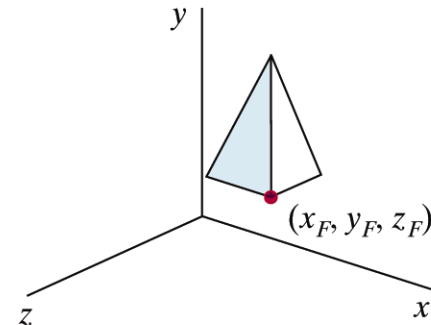
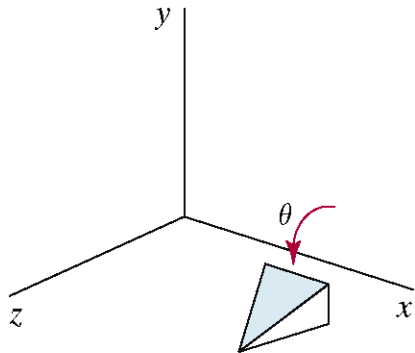
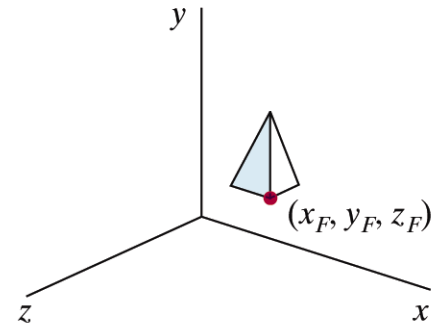
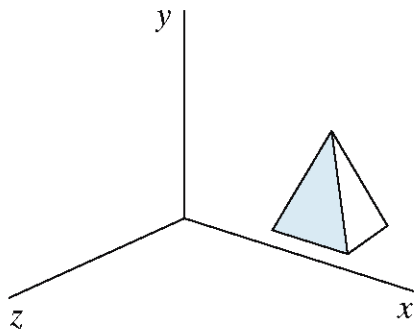
1. 3D Coordinate System
2. Vector Fundamentals
3. 3D Homogeneous Coordinate
- 4. 3D Geometric Transformations**
5. 3D Inverse Transformations
6. 3D Coordinate Transformations
7. Composite transformations

3D Geometric Transformations

- **Geometric Transformations:** In Geometric transformation an object itself is moved relative to a stationary coordinate system or background. The mathematical statement of this view point is described by geometric transformation applied to each point of the object. Various Geometric transformations are:
 - Translation
 - Scaling
 - Rotation
 - Reflection
 - Shearing

3D Geometric Transformations

- Once we have an object described, transformations are used to move that object, scale it and rotate it



Geometric Transformations

- Translation**

- Scaling

- Rotation

- Reflection

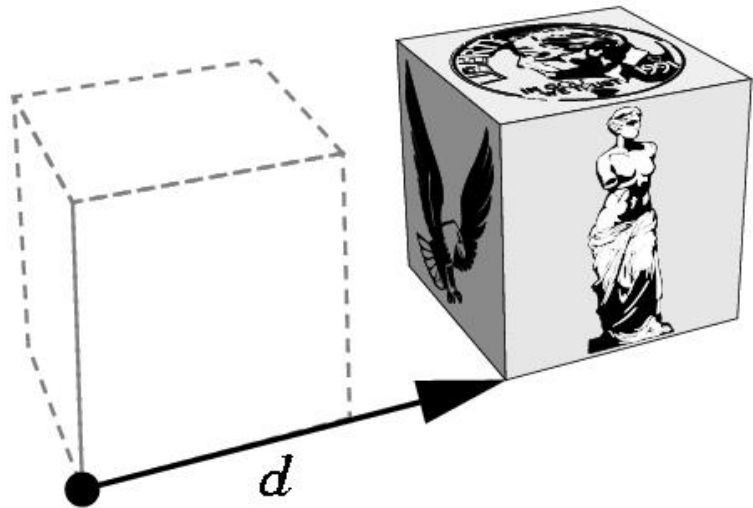
- Shearing

3D Geometric Translation

Translation is defined as the displacement of any object by a given distance and direction from its original position



(a)

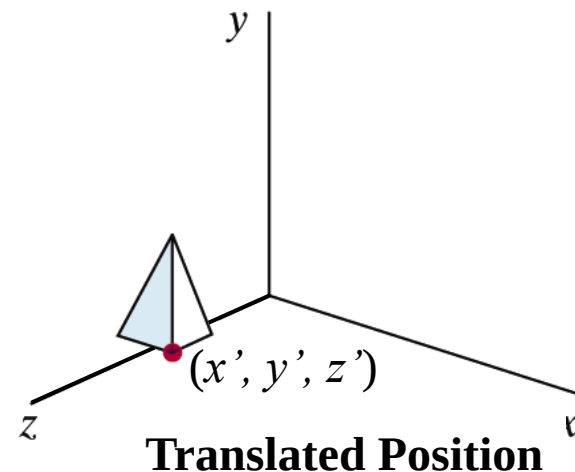
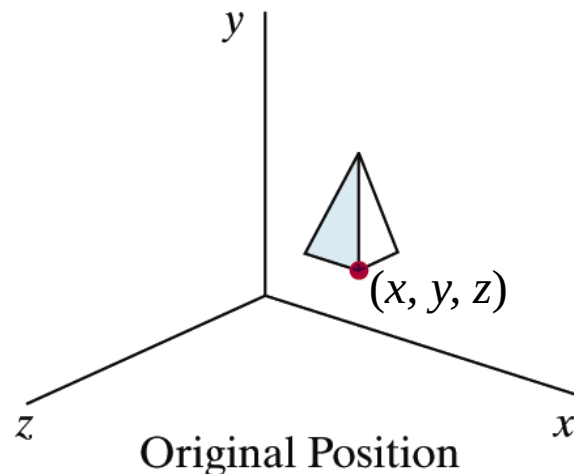


(b)

3D Geometric Translation

- To translate a point in three dimensions by tx , ty and tz simply calculate the new points as follows:

$$x' = x + tx \quad y' = y + ty \quad z' = z + tz$$



3D Geometric Translation

- The translation of a point $P(x,y,z)$ by (tx, ty, tz) can be written in matrix form as:

$$P' = T_v(P) \quad \text{where } v = tx\vec{I} + ty\vec{J} + tz\vec{K}$$

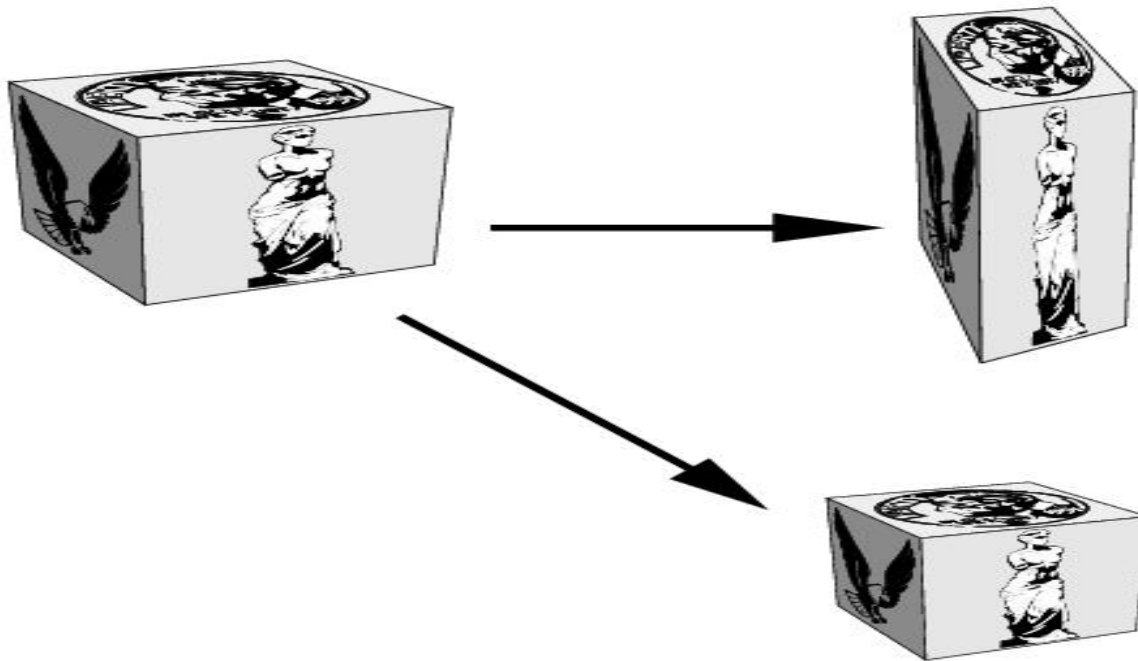
$$T_v = \begin{bmatrix} 1 & 0 & 0 & tx \\ 0 & 1 & 0 & ty \\ 0 & 0 & 1 & tz \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad P' = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} \quad P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Geometric Transformations

- Translation
- Scaling**
- Rotation
- Reflection
- Shearing

3D Geometric Scaling

Scaling is the process of expanding or compressing the dimensions of an object determined by the scaling factor.



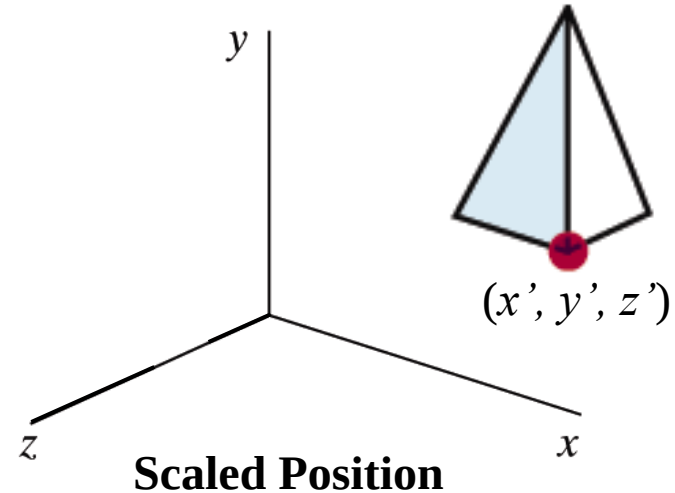
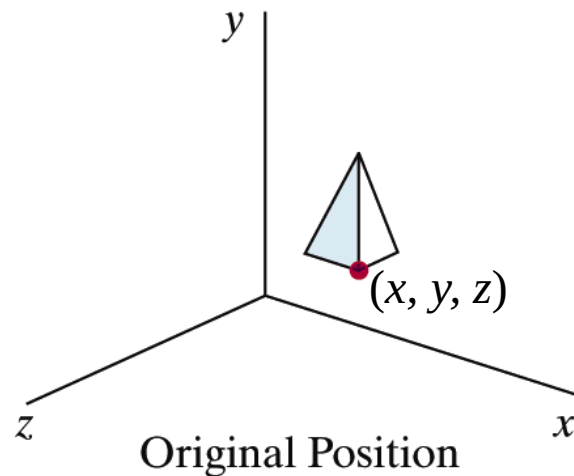
3D Geometric Scaling

- To scale a point in three dimensions by s_x , s_y and s_z simply calculate the new points as follows:

$$x' = s_x * x$$

$$y' = s_y * y$$

$$z' = s_z * z$$



3D Geometric Scaling

- The scaling of a point $P(x,y,z)$ by scaling factors S_x , S_y and S_z about origin can be written in matrix form as:

$$P' = S_{s_x, s_y, s_z}(P) \quad \text{where}$$

$$S_{s_x, s_y, s_z} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad P' = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} \quad P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

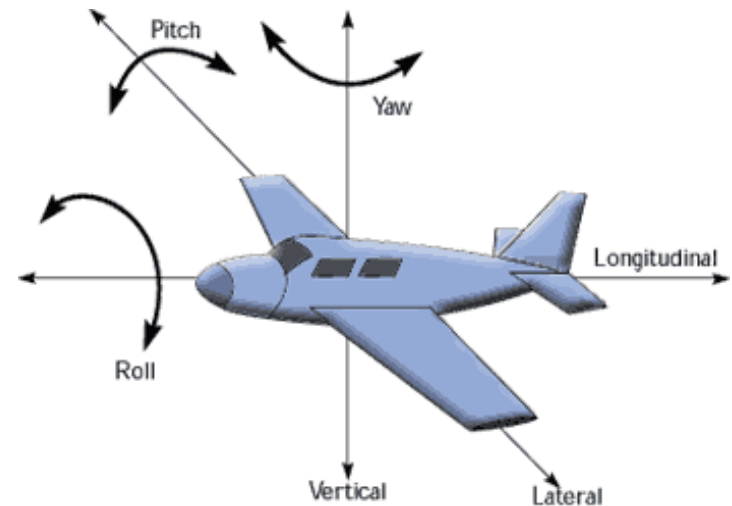
$$\text{such that} \quad \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} s_x \times x \\ s_y \times y \\ s_z \times z \\ 1 \end{bmatrix}$$

Geometric Transformations

- Translation
- Scaling
- Rotation**
- Reflection
- Shearing

3D Geometric Rotations

- When we performed rotations in two dimensions we only had the choice of rotating about the z axis
- In the case of three dimensions we have more options
 - Rotate about x – pitch
 - Rotate about y – yaw
 - Rotate about z - roll



3D Geometric Rotations

- **Roll** is known as the rising or dipping of the airplane's wing. The movement is done about **Longitudinal axis**. The Ailerons controlling the roll are located on the trailing edge of both wings.
- **Pitch** refers to the movement of the airplane's nose either up or down. Its movement is done about **Lateral axis**. The elevator controlling the pitch is also located on the rear of the aircraft on the tail, along with the rudder.
- **Yaw** allows the airplane to move towards the left or right while in flight. The movement is done about a **ventricle axis**. The yaw is controlled by the rudder located in the rear of the aircraft on the tail.



Roll



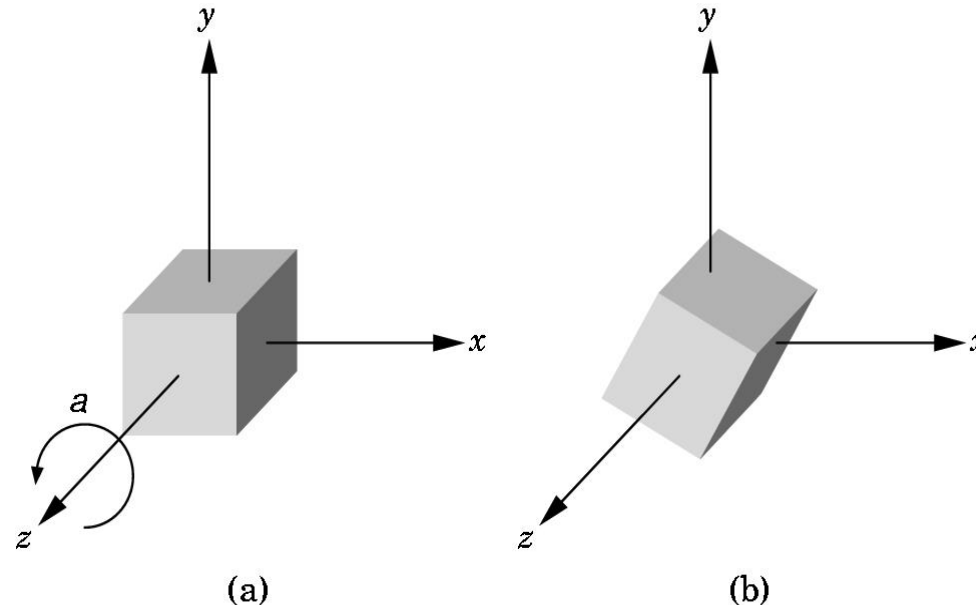
Pitch



Yaw

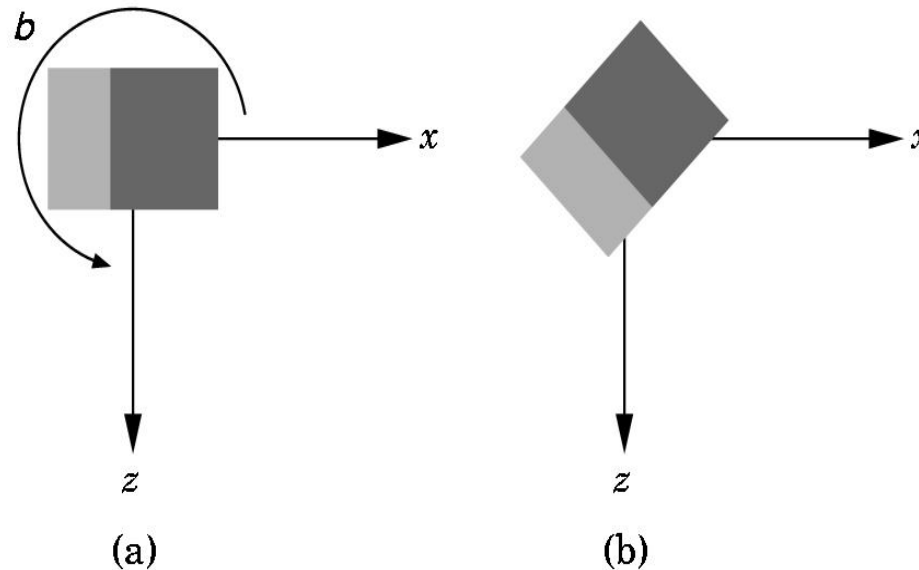
3D Geometric Rotations

- Rotate about z – axis: The picture shows a z -axis rotation around the origin in a positive angle, (Anti-clockwise) as you look down the z -axis towards the origin. The angle is measured in the xy -plane from the x -axis, just as in 2D.



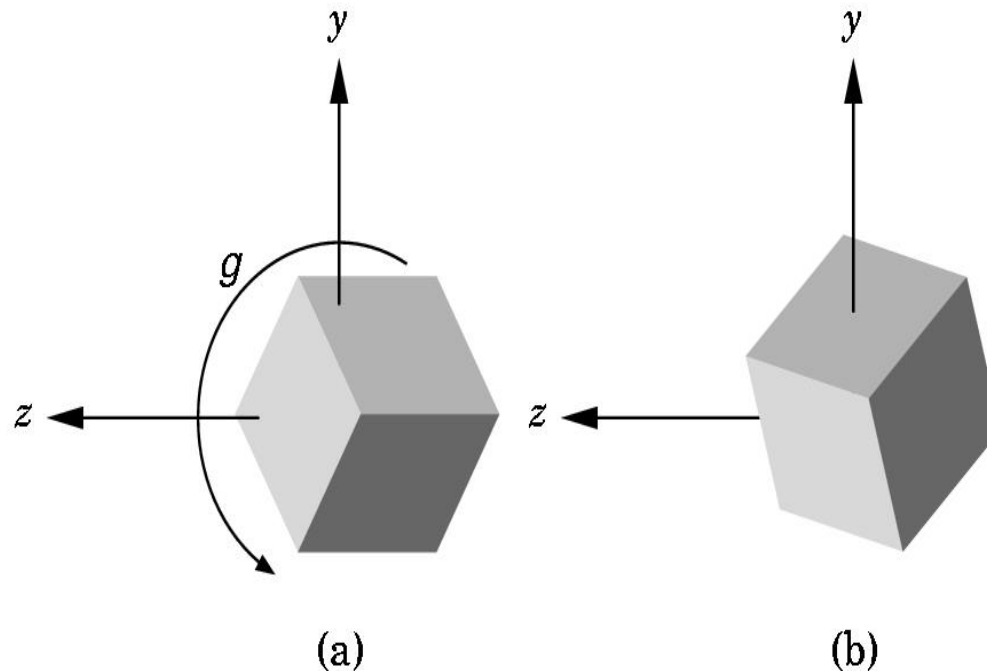
3D Geometric Rotations

- Rotate about y – axis: You are looking down the y -axis which is not shown. A positive (counter-clockwise) angle is shown.



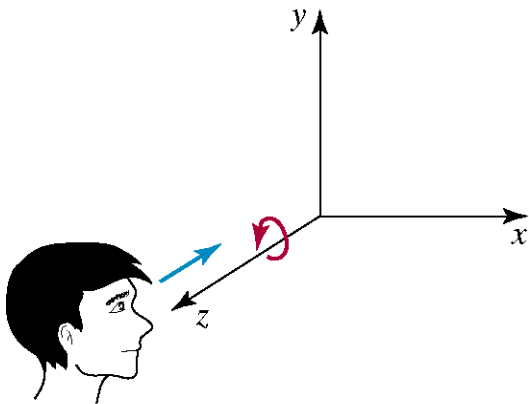
3D Geometric Rotations

- Rotate about x – axis: You are looking down the x -axis which is not shown. A positive (counter-clockwise) angle is shown.

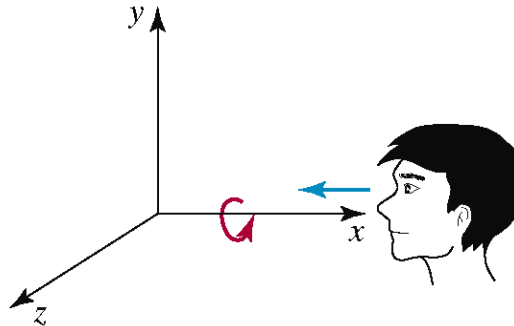


3D Geometric Rotations

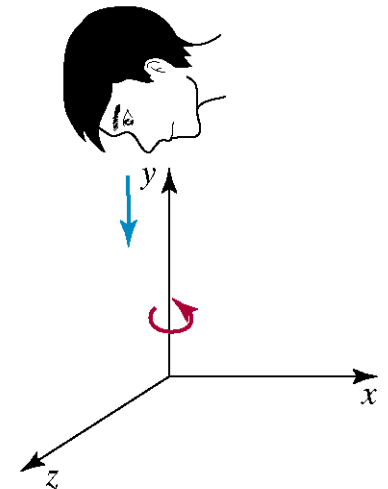
- The equations for the three kinds of rotations in 3-D are



$$\begin{aligned}x' &= x \cdot \cos\theta - y \cdot \sin\theta \\y' &= x \cdot \sin\theta + y \cdot \cos\theta \\z' &= z\end{aligned}$$



$$\begin{aligned}x' &= x \\y' &= y \cdot \cos\theta - z \cdot \sin\theta \\z' &= y \cdot \sin\theta + z \cdot \cos\theta\end{aligned}$$



$$\begin{aligned}x' &= z \cdot \sin\theta + x \cdot \cos\theta \\y' &= y \\z' &= z \cdot \cos\theta - x \cdot \sin\theta\end{aligned}$$

3D Geometric Rotations

- The scaling of a point $P(x,y,z)$ by an angle of θ about different axis about origin can be written in matrix form as:

$$P' = R_{\theta, axis}(P)$$

$$\text{where } P' = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} \quad P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad R_{\theta, K} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_{\theta, J} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_{\theta, I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Geometric Transformations

- Translation
- Scaling
- Rotation
- Reflection**
- Shearing

3D Geometric Reflections

- We can perform reflections relative to a selected *reflection axis* or with respect to a *reflection plane*.
- Reflections relative to a given axis are equivalent to 180° rotations about that axis.
- Reflections with respect to a plane are equivalent to 180° rotations in 4D space.
- 3 standard Reflections are
 - About z axis or with respect to xy plane
 - About y axis or with respect to zx plane
 - About x axis or with respect to yz plane

3D Geometric Reflections

- Reflection about XY – plane: Reflection relative to the xy plane can be obtained from following set of equations:

$$x' = x \qquad y' = y \qquad z' = -z$$

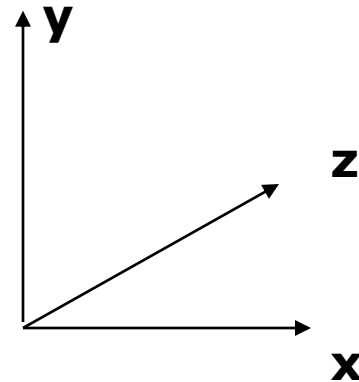
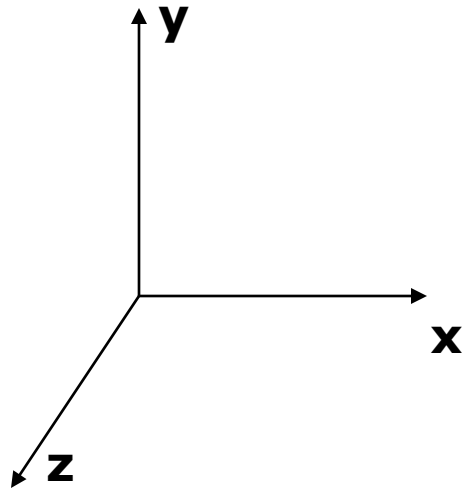
- In homogenous matrix form

$$P' = M_{xy}(P) \qquad \text{where} \qquad M_{xy} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Reflections about other planes can also be obtained by symmetry.

3D Geometric Reflections

- Reflection about XY – plane is a useful reflection as it converts a right-handed coordinate system into a left-handed coordinate system.



Contents

1. 3D Coordinate System
2. Vector Fundamentals
3. 3D Homogeneous Coordinate
4. 3D Geometric Transformations
- 5. 3D Inverse Transformations**
6. 3D Coordinate Transformations
7. Composite transformations

Inverse Transformations

- Inverse Translation: $T_v^{-1} = T_{-v}$
- Inverse Scaling: $S_{sx,sy,sz}^{-1} = S_{1/sx,1/sy,1/sz}$
- Inverse Rotations: $R_{\theta,p}^{-1} = R_{-\theta,p}$
- Inverse Reflections: $M_{plane}^{-1} = M_{plane}$

Contents

1. 3D Coordinate System
2. Vector Fundamentals
3. 3D Homogeneous Coordinate
4. 3D Geometric Transformations
5. 3D Inverse Transformations
- 6. 3D Coordinate Transformations**
7. Combining transformations

3D Coordinate Transformations

- **Coordinate Transformation:** The object is held stationary while coordinate system is moved relative to the object. These can easily be described in terms of the opposite operation performed by Geometric transformation.

3D Coordinate Transformations

- Coordinate Translation: $\overline{T}_v = T_v^{-1} = T_{-v}$
- Coordinate Scaling: $\overline{S}_{sx,sy,sz} = S_{sx,sy,sz}^{-1} = S_{1/sx,1/sy,1/sz}$
- Coordinate Rotations: $\overline{R}_{\theta,p} = R_{\theta,p}^{-1} = R_{-\theta,p}$
- Coordinate Reflections: $\overline{M}_{plane} = M_{plane}^{-1} = M_{plane}$

Contents

1. 3D Coordinate System
2. Vector Fundamentals
3. 3D Homogeneous Coordinate
4. 3D Geometric Transformations
5. 3D Inverse Transformations
6. 3D Coordinate Transformations
- 7. Composite transformations**

Composite Transformation

- Various composite transformations are
 1. Scaling about arbitrary point P
 2. Tilting as rotation about x-axis followed by y-axis
 3. Aligning vector V with K (Z – axis)
 4. Rotation about arbitrary axis L
 5. Mirror Reflection about arbitrary plane (N, R_0)

Exercise 1

Scaling about arbitrary point P

- Imagine scaling an object around a point (a,b,c) other than the origin
 - Translate point (a,b,c) to origin
 - Scale around origin
 - Translate back to point

Exercise 1

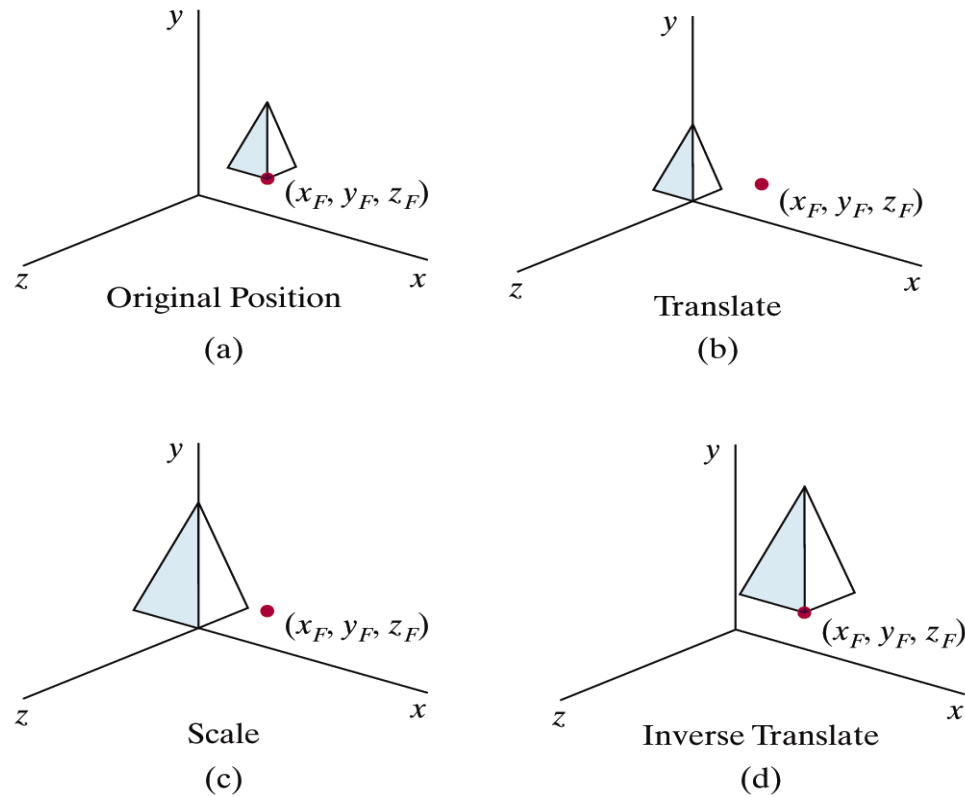


Figure 5-51

A sequence of transformations for scaling an object relative to a selected fixed point, using Eq. 5-110.

Exercise 1

Let P is the object point which is to be scaled by factors s_x , s_y and s_z about the fixed point (a,b,c) . Then the composite transformation $S_{s_x,s_y,s_z,(a,b,c)}$ can be obtained by performing following sequence of transformations :

1. Translate (a,b,c) to origin and the new object point is found as

$$P^1 = T_V(P) \text{ where } V = -aI - bJ - cK$$

2. Scale object about origin and the new object point is

$$P^2 = S_{s_x,s_y,s_z}(P^1)$$

3. Retranslate (a,b,c) back the final object point is

$$P^F = T_V^{-1}(P^2) = T_{-V}(P^2)$$

The composite transformation can be obtained by back substituting

$$P^F = T_V^{-1}(P^2)$$

$$= T_{-V} S_{s_x,s_y,s_z}(P^1)$$

$$= T_{-V} S_{s_x,s_y,s_z} \cdot T_V(P) \text{ where } V = -aI - bJ - cK$$

Exercise 1

Thus we form the matrix to be $S_{sx,sy,sz(a,b,c)} = T_V S_{sx,sy,sz} T_V$

And the composite scaling transformation matrix is

$$S_{sx,sy,sz(a,b,c)} = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} sx & 0 & 0 & 0 \\ 0 & sy & 0 & 0 \\ 0 & 0 & sz & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & -a \\ 0 & 1 & 0 & -b \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} sx & 0 & 0 & -a.sx + a \\ 0 & sy & 0 & -b.sy + b \\ 0 & 0 & sz & -c.sz + c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exercise 2

Define *tilting* as rotation about the x-axis followed by a rotation about y-axis.

- a) Find the tilting matrix $\mathbf{T} = \mathbf{R}_{\theta_y, \mathbf{J}} \times \mathbf{R}_{\theta_x, \mathbf{I}}$
- b) Does order of rotation matters ?

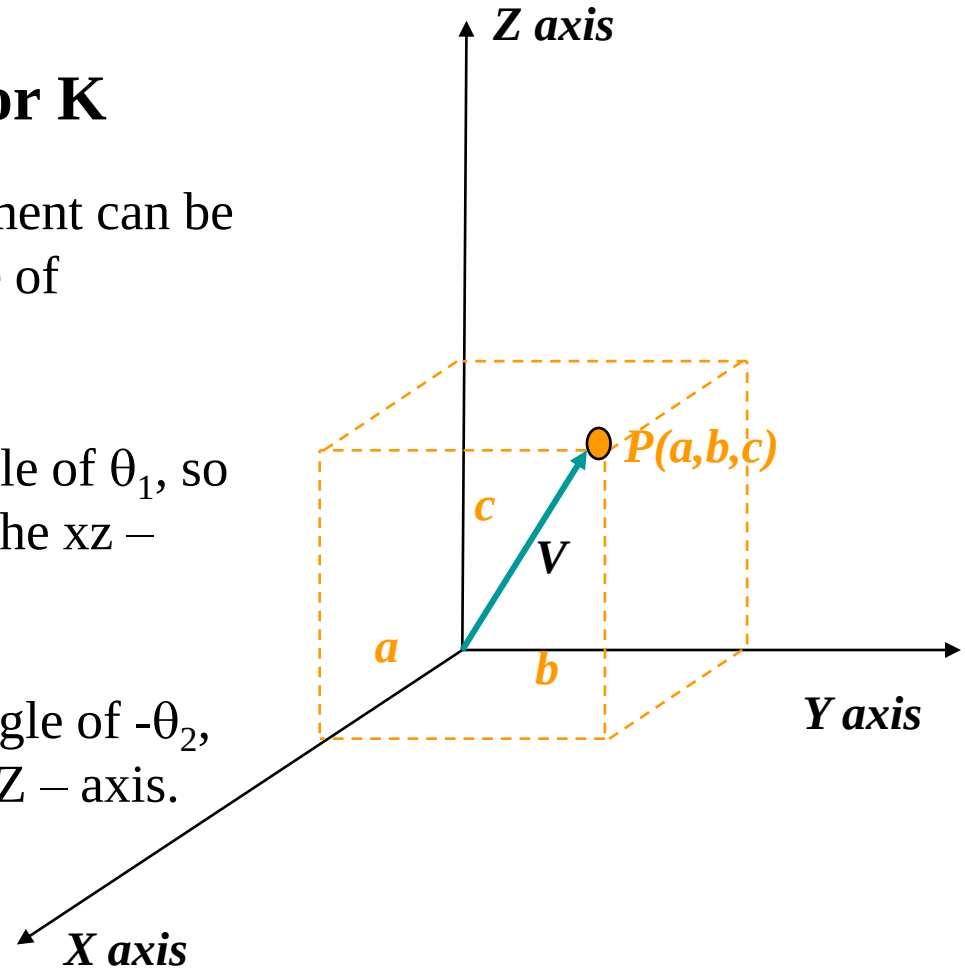
Exercise 3

Aligning vector V with Vector K

Let $V = aI + bJ + cK$ then the alignment can be performed by following sequence of operations:

- Rotate V about X -axis by an angle of θ_1 , so that V rotates into upper -half of the xz – plane. $V' = R_{\theta_1, I}(V)$
- Rotate V' about Y -axis by an angle of $-\theta_2$, so that V' rotates and align with Z – axis. $V'' = R_{-\theta_2, J}(V')$

How to get these two angles???



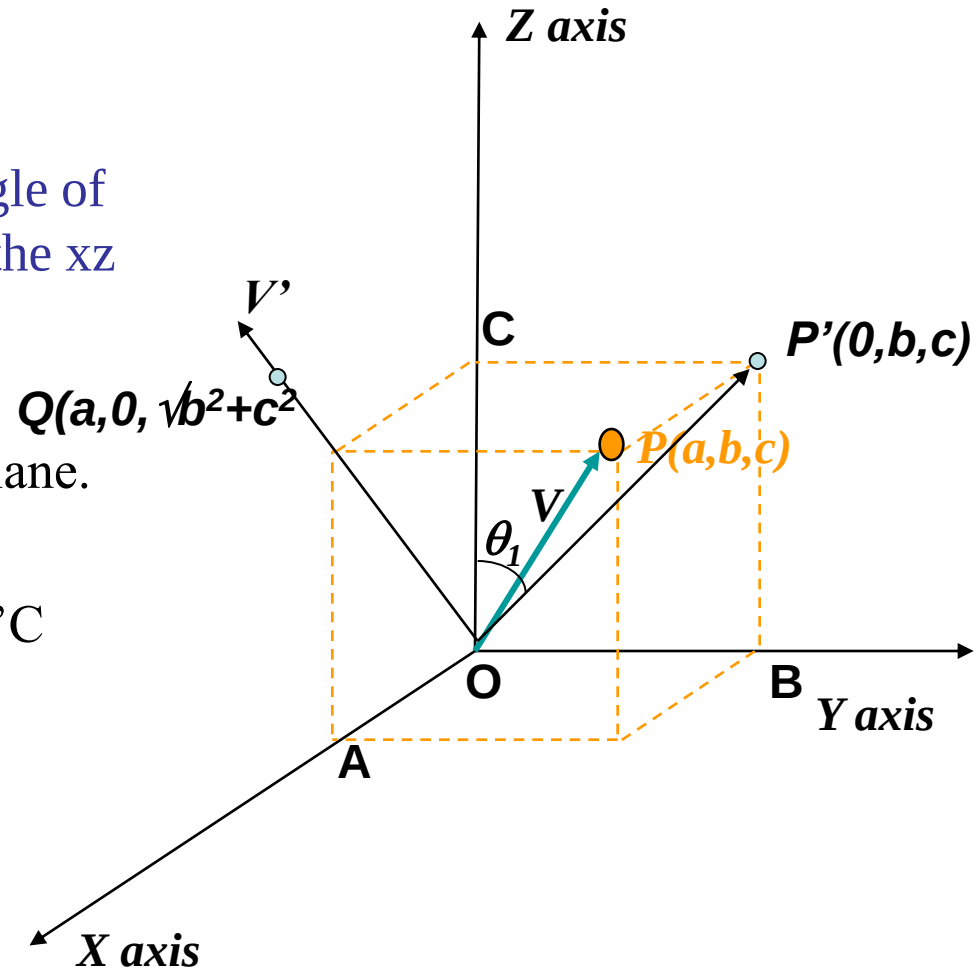
Exercise 3

Step1: Rotate V about X -axis by an angle of θ_1 , so that V rotates into upper -half of the xz - plane. $V' = R_{\theta_1, I}(V)$

Let P' be projection of $P(a,b,c)$ in YZ plane.
 $P' \leftrightarrow (0,b,c)$. If θ_1 be the angle that OP makes with +ve Z -axis, then from $\triangle OP'C$

$$\sin \theta_1 = \frac{b}{\sqrt{b^2 + c^2}}$$

$$\cos \theta_1 = \frac{c}{\sqrt{b^2 + c^2}}$$



Exercise 3

Rotate vector V about X -axis by an angle of θ_1

$$V' = R_{\theta_1, I}(V) \text{ or } Q = R_{\theta_1, I}(P)$$

Where

$$R_{\theta_1, I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 & 0 \\ 0 & \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\sin \theta_1 = \frac{b}{\sqrt{b^2 + c^2}}$$

$$\cos \theta_1 = \frac{c}{\sqrt{b^2 + c^2}}$$

Does V have rotated into upper -half of the xz – plane?

Exercise 3

$$Q = R_{\theta_1, I}(P) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 & 0 \\ 0 & \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{\sqrt{b^2 + c^2}} & -\frac{b}{\sqrt{b^2 + c^2}} & 0 \\ 0 & \frac{b}{\sqrt{b^2 + c^2}} & \frac{c}{\sqrt{b^2 + c^2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ \frac{cb - bc}{\sqrt{b^2 + c^2}} \\ \frac{b^2 + c^2}{\sqrt{b^2 + c^2}} \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} a \\ 0 \\ \sqrt{b^2 + c^2} \\ 1 \end{bmatrix}$$

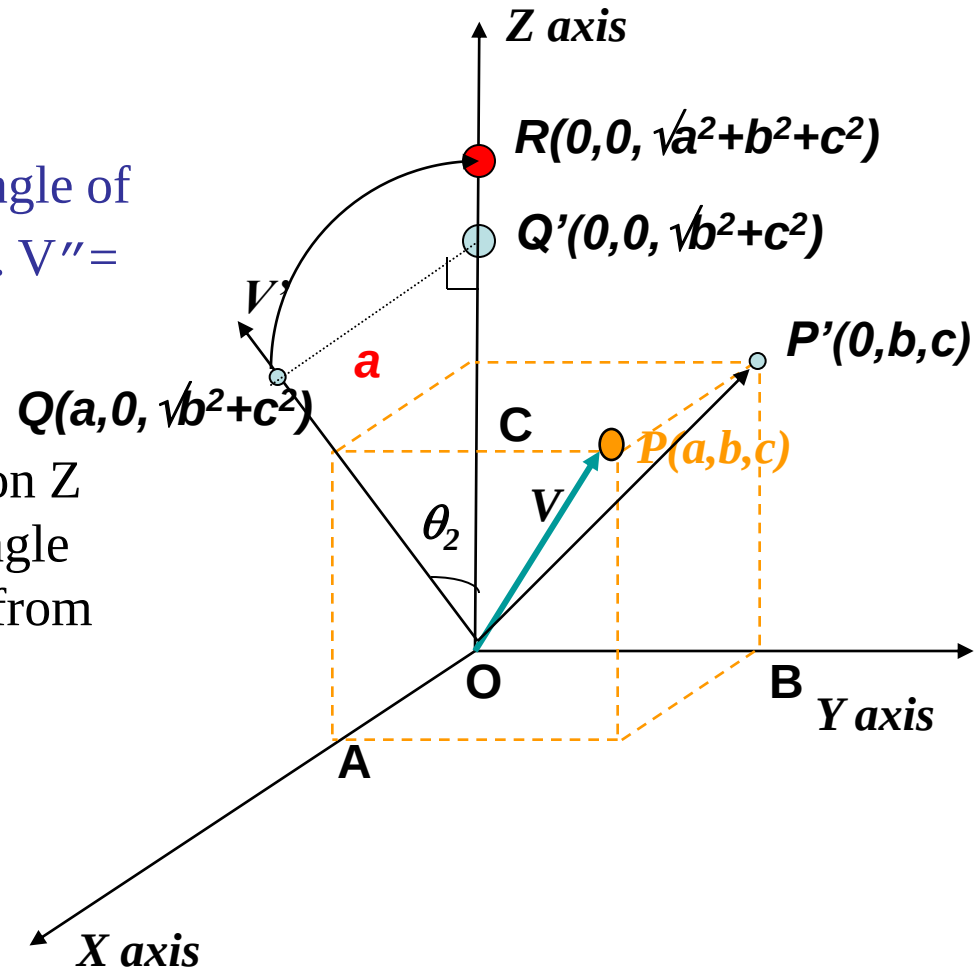
Exercise 3

Step2: Rotate V' about Y-axis by an angle of $-\theta_2$, so that V' aligns with +ve Z axis. $V'' = R_{\theta_1, I}(V)$

Let Q' be projection of $Q(a, 0, \sqrt{b^2 + c^2})$ on Z axis, $Q' \leftrightarrow (0, 0, \sqrt{b^2 + c^2})$. If θ_2 be the angle that OQ' makes with +ve Z-axis, then from $\Delta OQQ'$

$$\sin \theta_2 = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$$

$$\cos \theta_2 = \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}}$$



Exercise 3

Rotate vector V' about Y-axis by an angle of $-\theta_2$

$$V'' = R_{-\theta_2, J}(V') \text{ or } R = R_{-\theta_2, J}(Q)$$

Where

$$R_{-\theta_2, J} = \begin{bmatrix} \cos \theta_1 & 0 & -\sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ \sin \theta_2 & 0 & \cos \theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\sin \theta_2 = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$$
$$\cos \theta_2 = \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}}$$

Does V have aligned Z axis?

Exercise 3

$$\begin{aligned}
 R = R_{-\theta_2, J}(Q) &= \begin{bmatrix} \cos \theta_2 & 0 & -\sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ \sin \theta_2 & 0 & \cos \theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ 0 \\ \sqrt{b^2 + c^2} \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}} & 0 & -\frac{a}{\sqrt{a^2 + b^2 + c^2}} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{a}{\sqrt{a^2 + b^2 + c^2}} & 0 & \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ 0 \\ \sqrt{b^2 + c^2} \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{a\sqrt{b^2 + c^2} - a\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}} \\ 0 \\ \frac{a^2 + b^2 + c^2}{\sqrt{a^2 + b^2 + c^2}} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sqrt{a^2 + b^2 + c^2} \\ 1 \end{bmatrix}
 \end{aligned}$$

Exercise 3

Composite Matrix $A_{V,K}$ can be found as

$$R = R_{-\theta_2, J}(Q) = R_{-\theta_2, J} * R_{\theta_1, I}(P) =$$

$$= \begin{bmatrix} \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}} & 0 & -\frac{a}{\sqrt{a^2 + b^2 + c^2}} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{a}{\sqrt{a^2 + b^2 + c^2}} & 0 & \frac{\sqrt{b^2 + c^2}}{\sqrt{a^2 + b^2 + c^2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{0}{c} & -\frac{0}{b} & 0 \\ 0 & \frac{\sqrt{b^2 + c^2}}{b} & -\frac{\sqrt{b^2 + c^2}}{c} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\lambda}{|V|} & 0 & -\frac{a}{|V|} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{a}{|V|} & 0 & \frac{\lambda}{|V|} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{\lambda} & -\frac{b}{\lambda} & 0 \\ 0 & \frac{b}{\lambda} & \frac{c}{\lambda} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\lambda}{|V|} & \frac{-ab}{\lambda|V|} & -\frac{ac}{\lambda|V|} & 0 \\ 0 & \frac{c}{\lambda} & \frac{-b}{\lambda} & 0 \\ \frac{a}{|V|} & \frac{b}{|V|} & \frac{c}{|V|} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ where } \begin{matrix} \lambda = \sqrt{b^2 + c^2} \\ |V| = \sqrt{a^2 + b^2 + c^2} \end{matrix}$$

Exercise 3

Corollary I: If both $b = c = 0$

then $\lambda = 0$, $|V| = |a|$ and $\theta_2 = \pm 90^\circ$

$$A_{V,K} = \begin{bmatrix} 0 & 0 & -\frac{a}{|a|} & 0 \\ 0 & 1 & 0 & 0 \\ \frac{a}{|a|} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$
$$= R_{-\theta_2, J}$$

Exercise 3

Corollary II: The inverse of $A_{V,K}$ can be found as

$$\begin{aligned} A_{V,K}^{-1} &= (R_{-\theta_2,J} \cdot R_{\theta_1,I})^{-1} \\ &= R_{\theta_1,I}^{-1} \cdot R_{-\theta_2,J}^{-1} \\ &= R_{\theta_1,I}^T \cdot R_{-\theta_2,J}^T \\ &= (R_{-\theta_2,J} \cdot R_{\theta_1,I})^T \\ &= (A_{V,K})^T \\ &= A_{V,K}^T \end{aligned}$$

Exercise 4

Find transformation $A_{V,N}$ that aligns vector $V = 2I - J - K$ with another $N = I + J + K$.

$$A_{V,K} = \begin{bmatrix} \frac{\sqrt{2}}{\sqrt{6}} & 0 & \frac{2}{\sqrt{6}} & 0 \\ \frac{2}{\sqrt{2}\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 \\ \frac{2}{\sqrt{2}\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{K,N} = \begin{bmatrix} \frac{\sqrt{2}}{\sqrt{3}} & -\frac{1}{\sqrt{2}\sqrt{3}} & -\frac{1}{\sqrt{2}\sqrt{3}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

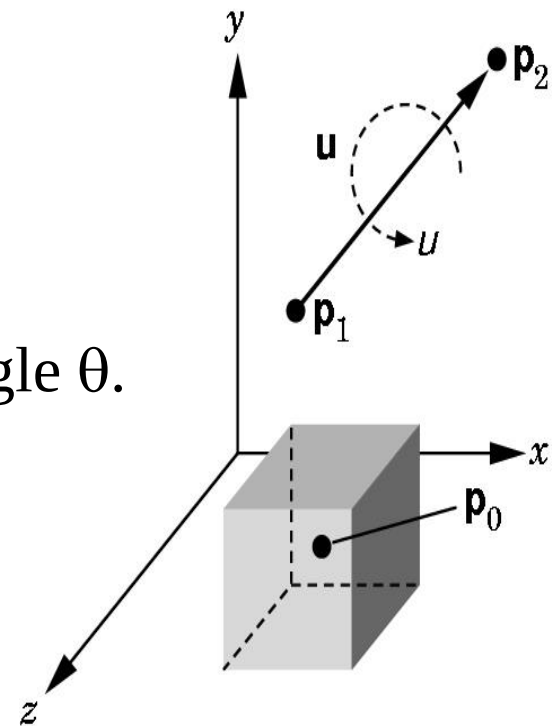
$$A_{V,N} = \begin{bmatrix} \frac{4}{\sqrt{18}} & \frac{1}{\sqrt{18}} & \frac{1}{\sqrt{18}} & 0 \\ \frac{1}{\sqrt{18}} & -\frac{2}{\sqrt{18}} - \frac{1}{2} & -\frac{2}{\sqrt{18}} + \frac{1}{2} & 0 \\ \frac{1}{\sqrt{18}} & -\frac{2}{\sqrt{18}} + \frac{1}{2} & -\frac{2}{\sqrt{18}} - \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exercise 5

Rotation by an angle θ about arbitrary axis L specified by a direction vector V and fixed point P_0 .

The transformation can be found as:

1. Translate P_0 to origin.
2. Align Vector V with K .
3. Rotate object about Z axis by angle θ .
4. Realign vector V back.
5. Retranslate P_0 back.



Exercise 5

If P is the object point the sequence of transformations is:

1. Translate P_0 to origin $\Rightarrow P^1 = T_{-P_0}(P)$
2. Align Vector V with K . $\Rightarrow P^2 = A_{V,K}(P^1)$
3. Rotate object about Z axis by angle θ . $\Rightarrow P^3 = R_{\theta,K}(P^2)$
4. Realign vector V back. $\Rightarrow P^4 = A_{V,K}^{-1}(P^3)$
5. Retranslate P_0 back. $\Rightarrow P^5 = T_{-P_0}^{-1}(P^4)$

$$\begin{aligned}\text{Thus } P^5 &= T_{-P_0}^{-1}(P^4) \\ &= T_{-P_0}^{-1} \cdot A_{V,K}^{-1}(P^3) \\ &= T_{-P_0}^{-1} \cdot A_{V,K}^{-1} \cdot R_{\theta,K}(P^2) \\ &= T_{-P_0}^{-1} \cdot A_{V,K}^{-1} \cdot R_{\theta,K} \cdot A_{V,K}(P^1) \\ &= T_{-P_0}^{-1} \cdot A_{V,K}^{-1} \cdot R_{\theta,K} \cdot A_{V,K} \cdot T_{-P_0}(P)\end{aligned}$$

Thus we form the matrix to be $R_{\theta,L} = T_{-P_0}^{-1} A_{V,K}^{-1} R_{\theta,K} A_{V,K} T_{-P_0}$

Exercise 5

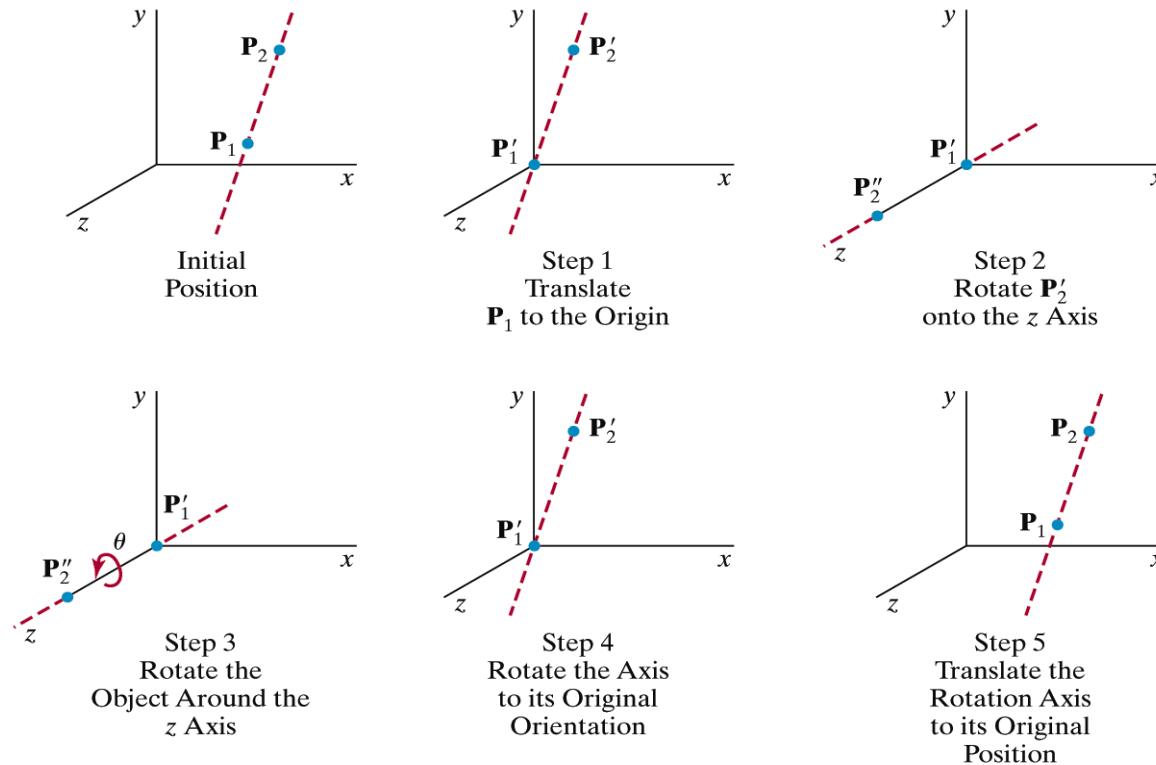


Figure 5-42

Five transformation steps for obtaining a composite matrix for rotation about an arbitrary axis, with the rotation axis projected onto the z axis.

Exercise 6

The pyramid is defined by A(0,0,0), B(1,0,0), C(0,1,0) and D(0,0,1) is rotated by 45° about line L in the direction of $V = J+K$ passing through P(0,1,0). Find the coordinates of rotated figure.

$$R_{45^\circ, L} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{2+\sqrt{2}}{4} & \frac{2-\sqrt{2}}{4} & \frac{2-\sqrt{2}}{4} \\ -\frac{1}{2} & \frac{2-\sqrt{2}}{4} & \frac{2+\sqrt{2}}{4} & \frac{2-\sqrt{2}}{4} \\ 0 & 0 & 0 & 1 \end{bmatrix} [A' B' C' D'] = \begin{bmatrix} \frac{1}{2} & \frac{1+\sqrt{2}}{2} & 0 & 1 \\ \frac{2-\sqrt{2}}{4} & \frac{4-\sqrt{2}}{4} & 1 & \frac{2-\sqrt{2}}{2} \\ \frac{\sqrt{2}-2}{4} & \frac{\sqrt{2}-4}{4} & 0 & \frac{\sqrt{2}}{2} \\ 1 & 1 & 1 & 1 \end{bmatrix}$$